A World Without Puzzle Pieces

Parthood, Scale and Life in Resolution Geometry

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Abstract

Education usually presents the world as a puzzle. Small pieces (quarks, atoms, cells) fit together into big pieces (organs, people, planets), and each piece is simply *there*, with a place and a size, whether or not anyone looks. This paper gives the alternative that Resolution Geometry supports. The alternative keeps every working result of anatomy, microbiology, chemistry and particle physics. What it changes is the meaning of "part".

A part is not a piece the world is made of.
It is a quotient that some protocol reads, and that closes on itself.

The argument rests on established results.

- *Parts are observable-induced.* The same state is maximally entangled for one choice of parts and a product for another (Zanardi; Zanardi–Lidar–Lloyd).
- *Wholes are not sums.* A deuteron weighs 2.2246 MeV less than its parts, and quark masses supply about 0.96% of the proton's mass.
- *Constituents depend on resolution.* The partons of a proton depend on the momentum transfer of the probe, and quarks are never isolated.
- *Elements are fibres.* Chemistry reads chlorine as one element (atomic weight 35.453, a mixture). Mass spectrometry splits it.
- *Scale separation is a ratio of clock counts,* not a separate realm. A slow variable closes with error $O(\varepsilon)$: 0.027, 0.0023, 0.00022 for $\varepsilon = 10^{-1}, 10^{-2}, 10^{-3}$.
- *Organisms are closures.* In a host–microbe model neither partner has a steady state alone, and together they do.

The answer to "is the bacterium inside the human?" Yes, for a position protocol. It is a separate organism, for its replication closure. It is part of a composite, for a joint metabolic closure. All three are true, and the puzzle picture's error is to demand one.

Limits. The paper adds no new force, particle or medical claim. It does not explain consciousness, and it treats claims about consciousness as protocol-relative attributions.

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1 What must be preserved, and what changes

Principle 1.1 (Preservation)**.** Every prediction that works must still work: anatomy, infection, chemistry, nuclear and particle physics. RG may change the *reading* of these results. It may not change their numbers.

Puzzle picture	RG picture	What stays the same in practice
The world is made of smallest pieces	The world is one; "pieces" are quotients read by protocols	every measured number
A part has a size and place in itself	Size and place are what a position protocol reads	microscopy, anatomy
The whole is the sum of its parts	Wholes carry interaction energy and closure the parts lack	binding energies, masses
Small and big are different realms	Small and big are resolutions of one world	Ehrenfest, coarse-graining
Time separates scales	Ratios of clock counts (delays) separate scales	rate constants, half-lives
An organism is a bounded piece	An organism is a closure relative to a task	microbiology, ecology

2 Parts are observable-induced

Definition 2.1 (Part)**.** Let a protocol family $\mathscr{P}$ read a system. A *part* for $\mathscr{P}$ is a factor A of a decomposition of the system's records, or of its observable algebra, $\mathcal{A} \cong \mathcal{A}_A \otimes \mathcal{A}_B$, such that the accessible interactions and measurements act locally on the factors. A *boundary* is a set of variables given which the inside and outside records are conditionally independent [3].

Imported Theorem 2.2 (Subsystems are induced by observables [1, 2])**.** *A Hilbert space of composite dimension admits many inequivalent tensor-product decompositions. Which decomposition counts as "the parts" is fixed by the algebra of accessible interactions and measurements, not by the space.*

Witness (P1). The state $(|00\rangle + |11\rangle)/\sqrt{2}$ has 1 ebit of entanglement between the usual two qubits. In the factorization obtained by the unitary $\mathrm{CNOT}(H \otimes I)$, which is a different choice of parts of the same four-dimensional space, it is a product state, with 0 ebit. Entanglement, separability and "being a part" are relative to the declared parts.

3 Wholes are not sums

Proposition 3.1 (Binding and confinement)**.**

- Nuclei. *The deuteron mass is $m_p + m_n - 2.2246\,MeV$ (P2). The whole weighs less than its pieces, and the difference is interaction energy.*
- Hadrons. *The current-quark masses u, u, d (2.16, 2.16, 4.67 MeV) supply about* 0.96% *of the proton mass* 938.27 *MeV. The rest is QCD interaction energy. The Higgs mechanism gives the quarks those small masses; it does not give the proton most of its mass.*
- Confinement. *Quarks are never observed in isolation. They are not removable pieces. They are labels of a structure that the proton's scattering response reveals.*

Imported Theorem 3.2 (Resolution-dependent constituents [4])**.** *In deep-inelastic scattering, the parton distribution functions of a proton evolve with the probe's momentum transfer Q^2 (DGLAP evolution). Probing at higher resolution reveals more quarks, antiquarks and gluons carrying smaller momentum fractions.*

Reading. Particle physics already has the RG answer: how many constituents a proton has is a resolution-dependent fact. "Atoms as balls" and "quarks as marbles" are puzzle images. The working theory is a theory of responses to probes.

4 Elements and isotopes: fibres of protocols

Proposition 4.1 (Element as a fibre)**.** *Chemistry, through spectra, bonding and valence, reads the nuclear charge Z, and it does not separate isotopes. Chlorine's atomic weight is $0.7576 \times 34.9689 + 0.2424 \times 36.9659 = 35.453$ (P3). It is a mixture inside one chemical fibre. A mass spectrometer is a finer protocol and splits it. Isotopes are* hidden *for chemistry and* visible *for mass spectrometry, exactly RG's typing of hidden values.*

The periodic table remains valid. It is the classification of the chemical protocol's fibres by Z.

5 Scale without "time as a thing"

Imported Theorem 5.1 (Tikhonov singular perturbation [6])**.** *For $\dot{x} = f(x, y)$ and $\varepsilon\dot{y} = g(x, y)$ with a stable slow manifold $g = 0$, the slow variable follows the reduced law $\dot{x} = f(x, h(x))$, up to $O(\varepsilon)$ after an initial layer.*

Witness (P4). For $\dot{x} = -x + y$ and $\varepsilon\dot{y} = -(y - x^2/(1 + x^2))$, the deviation from the reduced law is 0.0269, 0.00229 and 0.000225 for $\varepsilon = 10^{-1}$, 10^{-2} and 10^{-3}.
RG reading. Here ε is a *ratio* of relaxation counts: how many ticks of a declared clock the fast variable needs, relative to the slow one. Scale separation is a dimensionless ratio of delays, which in RG are counted on clocks. It requires no claim that time is a substance. Where those delays come from (path lengths, orbits, reaction barriers, transport) is physics. Their ratio is what lets a coarse description close.

6 Life: organisms as closures

Definition 6.1 (Closure; organism relative to a task)**.** A set of variables is *closed* for a task if its coarse dynamics evolves autonomously given only its boundary inputs. In RG terms, the time-derivatives of the coarse observables are functions of the coarse observables. An *organism* relative to a task is a maximal closed set that also maintains itself: under its boundary inputs it has a stable steady state or cycle.

Proposition 6.2 (Joint closure; P5)**.** *Consider a host H that needs a vitamin V made only by a microbe M, and a microbe that needs a nutrient N supplied only through the host.*

- *Alone, each has negative growth ($-\frac{1}{2}$ and $-\frac{2}{5}$), so neither has a positive steady state.*
- *Together they close, with $(H, M, V, N) = (\frac{25}{72}, \frac{3}{8}, 1, \frac{2}{3})$.*

The bacterium question, answered.

- *Position protocol:* the bacterium is inside the human, as microscopy and anatomy correctly say.
- *Replication closure:* it is a separate organism. It has its own genome, it can be cultured, and it closes its own cycle given nutrients.
- *Joint metabolic closure,* for example fibre digestion or vitamin synthesis: host and microbiota form one closed composite (the holobiont) for that task.

These are three valid quotients of one world. The puzzle picture forces a single decomposition. RG keeps all three and says which protocol each belongs to.

Viruses and transfer. A virus does not close its replication alone; it closes only with a host's machinery. "Is a virus alive?" is therefore a question about which closure is declared, not about a hidden essence. Transmission is mediation along a contact network. The wall-tap theorems apply: no test detects infection before the mediated arrival, the optimal error before arrival is $\frac{1}{2}$, and detection follows distinguishability, not exposure. Illness, in RG terms, is:

- a loss of closure, when a tissue's coarse dynamics stops being autonomous;
- a change of transfer function, a different response to the same inputs;
- or an approach to a wall, a bifurcation preceded by critical slowing down: rising autocorrelation and variance.

These are diagnostic categories, not medical claims.

Water and life. The anomalous properties of liquid water (density maximum, high heat capacity, hydrogen-bond networks) are properties of an interacting whole, read by thermodynamic protocols. Life is a driven, non-equilibrium closure. It persists because energy flows through it and it continually re-imposes its own structure, for example by proofreading [7, 8]. The H-theorem's monotone approach to equilibrium applies only when the dynamics fixes the equilibrium, and a living cell is maintained away from that condition.

7 Small and big at the same time

"Small" and "big" are resolutions of one world, not places within it. The quantum law is the single law. What depends on scale is which of its features a protocol can read.

- A fine protocol reads interference and Leggett–Garg violations.
- A coarse protocol reads classical means. For Lie-algebraic dynamics these follow the classical law exactly (One World paper).

So quantum mechanics is not "just a scale of measurement". It is the law, and quantum *effects* are what fine protocols see of it. The world is small and big at the same time because one state is read at many resolutions, not because it is assembled from small pieces.

8 Consciousness: what RG can and cannot say

- **What RG can type.** Attributions of consciousness rest on protocols: reports, behaviour, and measurable correlates such as the perturbational complexity index [9]. Each is a fibre-forming protocol, with hidden and null directions.
- **What RG cannot claim.** RG does not explain why there is experience (the hard problem [10]). It gives no evidence of "micro-consciousness" and does not endorse panpsychism. "Consciousness of a bacterium" is, at most, a question about which closure and which protocol is declared. RG supplies no measurement that answers it.

9 Ledger

Claim	Status	Basis
Parts are induced by accessible observables	IMPORTED + EXACT WITNESS	Zanardi; ZLL; P1
Wholes carry energy the parts lack	COMPUTED	deuteron; proton (P2)
Constituents depend on probe resolution	IMPORTED	DGLAP, confinement
Elements are chemical fibres; isotopes hidden	COMPUTED	P3
Scale separation is a ratio of clock counts	IMPORTED + EXECUTED	Tikhonov; P4
Organism = closure relative to a task; several valid at once	TOY PROOF	P5
Infection obeys mediation and testing bounds	THEOREM (Wall-Tap Structural)	delay paper
Quantum effects are what fine protocols read of one law	CONDITIONAL	One World paper
Explanation of consciousness	NOT CLAIMED	—
Medical diagnosis or treatment	NOT CLAIMED	needs data and validation

10 Reproduction

`rg_world_witnesses.py` runs P1–P5, deterministically, and reruns are byte-identical.

- P1: exact entanglement of one state under two choices of parts.
- P2, P3: published masses and isotope data.
- P4: a Radau integration of the fast–slow system.
- P5: exact algebra for the host–microbe model.

The host–microbe network is a toy model, not physiology.

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